Article

Robust Tracking and Stabilization of Remote-Controlled Weapon Station Systems

Chhavi Singh

[1] Indian Institute of Technology Madras, Chennai 600036, India; chhavi.akan@gmail.com

Abstract

Line-of-sight (LOS) stabilization in remote-controlled weapon stations (RCWS) is challenged by nonlinear cross-axis inertial coupling, dynamic gravity torques and multi-frequency hull disturbances. This paper presents a multi-rate cascaded architecture for precision LOS tracking under realistic vehicle kinematics and sensor non-idealities. The architecture combines a 1 kHz outer position loop, a 10 kHz kinematic feedforward compensator and a 20 kHz inner Sliding Mode Control-Active Disturbance Rejection Control (SMC-ADRC) loop with an Extended State Observer. Full analytical spatial coupling dynamics derived from Euler's rigid-body formulation are incorporated into the feedforward path for anticipatory disturbance compensation. The controller is validated on a high-fidelity, CAD-derived Simscape Multibody digital twin incorporating realistic IMU noise, including angle random walk (ARW), rate random walk (RRW) and bias instability. Under the modeled disturbance and sensor conditions, the proposed architecture achieves steady-state pointing errors of 0.118 mrad RMS in azimuth and 0.544 mrad RMS in elevation, demonstrating sub-milliradian LOS stabilization.

Keywords: Robust Control; Inertial Stabilized Platform; Sliding Mode Control; Disturbance Observer

1. Introduction

During combat engagement, a vehicle-mounted remote-controlled weapon system (RCWS) must maintain precise line-of-sight (LOS) tracking despite vehicle-induced disturbances. Roll, pitch, yaw and heave motions of the vehicle can significantly degrade turret pointing accuracy. This work presents a multi-rate cascaded architecture for precision LOS tracking under realistic vehicle kinematics and sensor non-idealities.

2. Methodology

2.1. System Architecture

The overall control architecture is shown in Fig. 1. A cascaded, multi-rate structure is employed for LOS tracking and disturbance compensation. The outer Proportional-Integral (PI) position loop operates at 1 kHz and generates azimuth and elevation rate commands from the LOS tracking errors. A kinematic feedforward compensator operates at 10 kHz, using measured vehicle angular rates to generate anticipatory counter-rate commands. These commands are combined with the PI output and supplied to the inner torque-control loop.

The inner loop operates at 20 kHz ($T_s = 50\,\mu$s) and employs a Sliding Mode Controller (SMC) with an Extended State Observer (ESO). The SMC provides robust velocity tracking against model uncertainties and nonlinearities, while the ESO estimates the lumped disturbance for explicit compensation within the control law. The resulting torque command drives the RCWS plant, with measured angular position and rate fed back to the respective control loops, as shown in Fig. 1.

The system is evaluated under representative roll, pitch, and yaw hull motions generated using prescribed sinusoidal profiles. The corresponding angular rates are supplied to the feedforward path. Sensor non-idealities include stochastic gyro errors characterized by Angle Random Walk (ARW), Rate

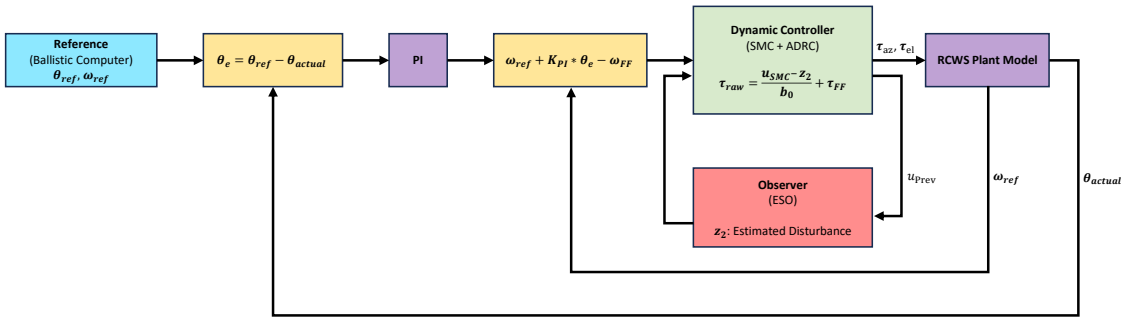


Figure 1. Multi-rate cascaded control architecture for RCWS LOS stabilization.

Random Walk (RRW), and slowly varying bias instability, along with encoder noise and measurement errors.

2.2. Dynamic Model

The rotational dynamics of the two-axis RCWS are derived from Euler's rigid-body equation. The spatial coupling formulation follows the derivation presented in [1]. The inertia matrices of the pitch and yaw gimbals are respectively defined as

$$\mathbf{J}_a = \begin{bmatrix} J_{ax} & D_{xy} & D_{xz} \\ D_{xy} & J_{ay} & D_{yz} \\ D_{xz} & D_{yz} & J_{az} \end{bmatrix}, \qquad \mathbf{J}_k = \begin{bmatrix} J_{kx} & d_{xy} & d_{xz} \\ d_{xy} & J_{ky} & d_{yz} \\ d_{xz} & d_{yz} & J_{kz} \end{bmatrix}. \tag{1}$$

where J_{ii} denote the principal moments of inertia, while D_{ij} and d_{ij} denote the products of inertia of the pitch and yaw assemblies, respectively.

2.2.1. Azimuthal Channel Dynamics

Let the absolute angular velocity of the azimuth frame k be $\boldsymbol{\omega}_k = \begin{bmatrix} p_k & q_k & r_k \end{bmatrix}^T$, where p_k, q_k, and r_k are the angular-rate components expressed in the azimuth frame.

The yaw-axis dynamics can be expressed as

$$J_k(\theta_2)\dot{r}_k = T_z + T_{d1} + T_{d2} + T_{d3}, \tag{2}$$

where T_z is the external torque about the azimuth axis, θ_2 is the relative elevation angular position, $J_k(\theta_2)$ is the effective azimuthal inertia and T_{d1}, T_{d2}, and T_{d3} represent gyroscopic, inertia-asymmetry, and configuration-dependent coupling terms, respectively.

The effective azimuthal inertia is obtained from the transformed payload inertia as

$$J_k(\theta_2) = J_{kz} + J_{ax}\sin^2\theta_2 + J_{az}\cos^2\theta_2 - D_{xz}\sin(2\theta_2). \tag{3}$$

The gyroscopic coupling term is

$$T_{d1} = \Big(J_{x,\mathrm{eff}}(\theta_2) - J_{y,\mathrm{eff}}(\theta_2)\Big)p_k q_k, \tag{4}$$

while the product-of-inertia coupling is

$$\begin{aligned} T_{d2} = & -D_{xz,\mathrm{eff}}(\theta_2)(\dot{p}_k - q_k r_k) \\ & - D_{yz,\mathrm{eff}}(\theta_2)(\dot{q}_k + p_k r_k) - D_{xy}\Big(p_k^2 - q_k^2\Big). \end{aligned} \tag{5}$$

$$
\begin{aligned}
T_{d3} = \ddot{\theta}_2 &\left(D_{xy}\sin\theta_2 - D_{yz}\cos\theta_2\right) \\
&+ \dot{\theta}_2\Big[(J_{ax} - J_{az})p_k\cos(2\theta_2) - r_k\sin(2\theta_2) \\
&\quad + 2D_{xz}(p_k\sin(2\theta_2) + r_k\cos(2\theta_2)) \\
&\quad + (D_{yz}\sin\theta_2 + D_{xy}\cos\theta_2)(q_a + q_k) - J_{ay}p_k\Big].
\end{aligned} \tag{6}
$$

2.2.2. Elevation Channel Dynamics

Let the absolute angular velocity of the elevation frame a be $\boldsymbol{\omega}_a = \begin{bmatrix} p_a & q_a & r_a \end{bmatrix}^T$, where p_a, q_a, and r_a are the angular-rate components expressed in the elevation frame. The resulting equation is

$$
J_{ay}\dot{q}_a = T_y + T_B + T_C \tag{7}
$$

where q_a is the elevation angular velocity and T_y is the total external torque applied about the elevation axis.

Following the frame transformation in [**?**], the coupling disturbance can be decomposed as

$$
T_D = T_B + T_C, \tag{8}
$$

where T_B represents the coupling induced by vehicle/body motion:

$$
\begin{aligned}
T_B = &-(D_{yz}\sin\theta_2 + D_{xy}\cos\theta_2)(\dot{p}_k + q_k r_k) \\
&+ (D_{yz}\cos\theta_2 - D_{xy}\sin\theta_2)p_k q_k \\
&+ [(J_{az} - J_{ax})\cos(2\theta_2) - 2D_{xz}\sin(2\theta_2)]p_k r_k \\
&+ \frac{1}{2}[(J_{az} - J_{ax})\sin(2\theta_2) + 2D_{xz}\cos(2\theta_2)]p_k^2,
\end{aligned} \tag{9}
$$

and T_C represents the coupling associated with the yaw-rate motion:

$$
\begin{aligned}
T_C = &(D_{xy}\sin\theta_2 - D_{yz}\cos\theta_2)\dot{r}_k \\
&- \frac{1}{2}[(J_{az} - J_{ax})\sin(2\theta_2) + 2D_{xz}\cos(2\theta_2)]r_k^2.
\end{aligned} \tag{10}
$$

2.2.3. Gravity Torque

The static center-of-gravity offset of the elevation assembly introduces an additional gravitational disturbance torque. The gravitational torque about the elevation axis is modeled as

$$
\tau_g(\theta_2) = mgr_\perp\cos(\theta_2 + \phi_0), \tag{11}
$$

where m is the payload mass, g is the gravitational acceleration, and $r_\perp$ is the perpendicular distance from the elevation axis to the center of gravity. ϕ_0 is corresponding static angular offset.

Therefore, the total elevation-axis torque can be represented as

$$
T_y = \tau_{\text{motor}} + \tau_g + \tau_{\text{ext}}, \tag{12}
$$

where τ_{motor} is the actuator torque and τ_{ext} represents other external disturbance torques.

2.3. Control Architecture

2.3.1. Reference Tracking and Feedforward Compensation

The position tracking error is defined as

$$e_\theta(k) = \theta_{ref}(k) - \theta_{act}(k). \tag{13}$$

The nominal angular-velocity command generated by the discrete PI controller is

$$\omega_{PI}(k) = K_p e_\theta(k) + K_i \int_0^t e_\theta(\tau)\, d\tau + \omega_{ref} \tag{14}$$

where $K_p = 37$ and $K_i = 35$.

The kinematic feedforward compensation for vehicle motion is

$$\omega_{ff} = \begin{bmatrix} -r \\ p \sin\theta_1 - q \cos\theta_1 \end{bmatrix}, \tag{15}$$

where p, q, and r are the hull angular rates measured by the vehicle-mounted IMU. The compensated velocity reference is therefore

$$\omega_{comp} = \omega_{PI} + \omega_{ff}. \tag{16}$$

2.3.2. Extended State Observer

The ESO estimates the actual angular velocity and the lumped unknown disturbance. The observer error is

$$e_{eso} = z_1 - \omega_{act}, \tag{17}$$

with observer dynamics

$$\dot{z}_1 = z_2 + b_0 u_{prev} - \beta_1 e_{eso}, \tag{18}$$

$$\dot{z}_2 = -\beta_2 e_{eso}, \tag{19}$$

where z_1 is the estimated angular velocity and ω_{act} is the measured angular velocity, $b_0 = 1/J$ and z_2 denotes the lumped disturbance estimate. For an observer bandwidth $\omega_o = 350$ rad/s,

$$\beta_1 = 2\omega_o, \qquad \beta_2 = \omega_o^2. \tag{20}$$

2.3.3. Sliding Mode Controller

The velocity tracking error and sliding surface are defined as

$$e_v = \omega_{comp} - \omega_{act}, \qquad s = e_v. \tag{21}$$

To reduce chattering, the discontinuous switching function is replaced by a saturation function,

$$\mathrm{sat}\Big(\frac{s}{\Phi}\Big) = \begin{cases} 1, & s > \Phi, \\ s/\Phi, & |s| \le \Phi, \\ -1, & s < -\Phi. \end{cases} \tag{22}$$

The SMC control action is

$$u_{smc} = \dot{\omega}_{comp} + c\, e_v + K\, \mathrm{sat}\Big(\frac{s}{\Phi}\Big), \tag{23}$$

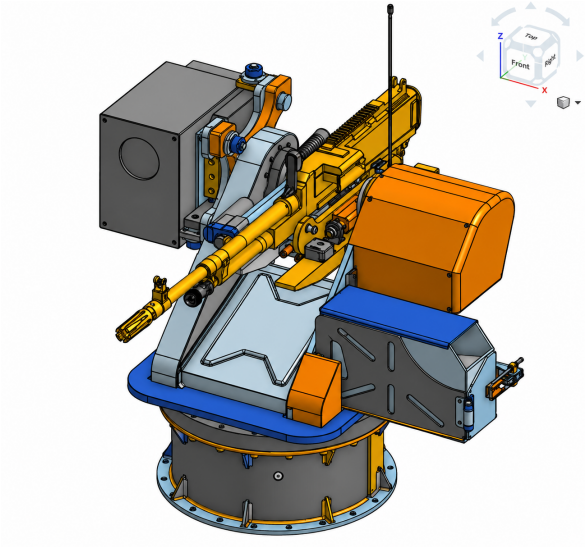
where $c = 180$, $K = 60$, and $\Phi = 0.08$.

2.3.4. Total Control Law

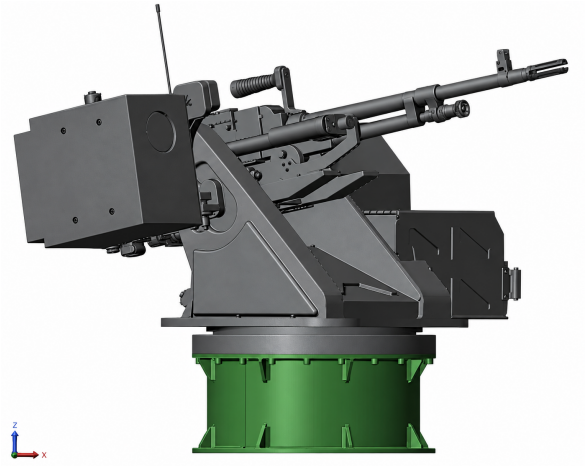
The final torque command combines the robust SMC action with ESO-based disturbance compensation and known dynamic feedforward terms:

$$\tau_{raw} = \frac{u_{smc} - z_2}{b_0} + \tau_{known}, \tag{24}$$

where τ_{known} represents explicitly modelled effects such as gravity and other known dynamic terms . The resulting torque command is finally limited according to the physical actuator constraints.



(**a**) CAD model of the RCWS



(**b**) CAD-derived Simscape Multibody model of the RCWS

3. Experimental Setup

A high-fidelity RCWS plant was developed in Simscape Multibody from CAD-derived geometry and represented in URDF, incorporating rigid-body geometry, mass properties, centers of gravity, and inertia tensors. The CAD and corresponding Simscape models are shown in Figs. **??** and **??**, respectively. Aluminium 6061, with a density of 2.71 g/cm^3, was used for the CAD model. The model captures the mechanical dynamics, actuator behavior, and nonlinearities of the RCWS. The principal system parameters are listed in Table 1.

The RCWS base was connected through a 3-DOF Gimbal Joint to impose roll, pitch, and yaw disturbances. Sinusoidal hull motions with amplitudes of 0.2618, 0.0873, and 0.1745 rad and frequencies

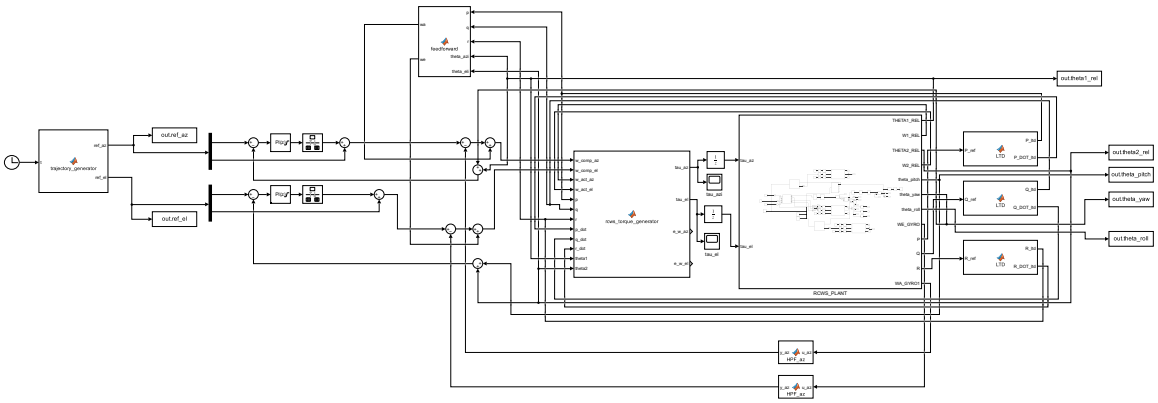


Figure 2. Simulink controller architecture.

of 0.1048, 0.0345, and 0.0524 rad/s were applied in roll, pitch, and yaw, respectively. Joint friction and viscous damping were included to capture mechanical nonlinearities.

A tactical-grade MEMS IMU was modelled to provide hull angular rates p, q, and r, incorporating Angle Random Walk (ARW), Rate Random Walk (RRW), and bias instability through double-sided noise models. Independent gyroscopes for the azimuth and elevation axes were similarly modelled with representative ARW, RRW, and bias-instability characteristics to assess robustness to measurement uncertainties. The implemented Simulink controller architecture is shown in Fig. 2.

Table 1. RCWS plant parameters.

Parameter	**Value**	**Unit**
CAD material	Aluminium 6061	–
Density	2.71	$\mathrm{g/cm^3}$
Gravitational acceleration, g	9.81	$\mathrm{m/s^2}$
Elevation mass, m_e	25.1892	kg
Azimuth mass, m_a	17.252959	kg
Perpendicular distance, $r_\perp$	0.206155	m
Initial elevation angle, ϕ_0	0.244979	rad
ARW of Hull IMU	0.06	$^\circ/\sqrt{\mathrm{hr}}$
Bias Instability of Hull IMU	0.45	$^\circ/\mathrm{hr}$
RRW of Hull IMU	0.01	$^\circ/\sqrt{\mathrm{hr}}$
Azimuth inertia tensor, J_k ($\mathrm{kg\,m^2}$)		
$\begin{bmatrix} 5.7232765 & -0.32806335 & -1.9506277 \\ -0.32806335 & 6.0696006 & -0.27789085 \\ -1.9506277 & -0.27789085 & 4.1196229 \end{bmatrix}$		
Elevation inertia tensor, J_a ($\mathrm{kg\,m^2}$)		
$\begin{bmatrix} 1.8626912 & 0.17617059 & 0.055107026 \\ 0.17617059 & 1.0532286 & 0.051133846 \\ 0.055107026 & 0.051133846 & 2.6931045 \end{bmatrix}$		
Azimuth COG, (x, y, z) (m)		
$(0.000223,\ -0.006411,\ 0.108807)$		
Elevation COG, (x, y, z) (m)		
$(-0.038850,\ 0.190150,\ -0.008000)$		

4. Results

The reference commands for the azimuth and elevation axes are set to target angles of 0.2 rad and 0.1 rad, respectively, corresponding to the desired pointing commands generated by the ballistic computer. As shown in Fig. 5, the proposed control architecture achieves steady-state pointing RMSEs of 0.118 mrad for azimuth and 0.544 mrad for elevation, satisfying the specified requirement of less than 1 mrad.**Demo video can be accessed here.**

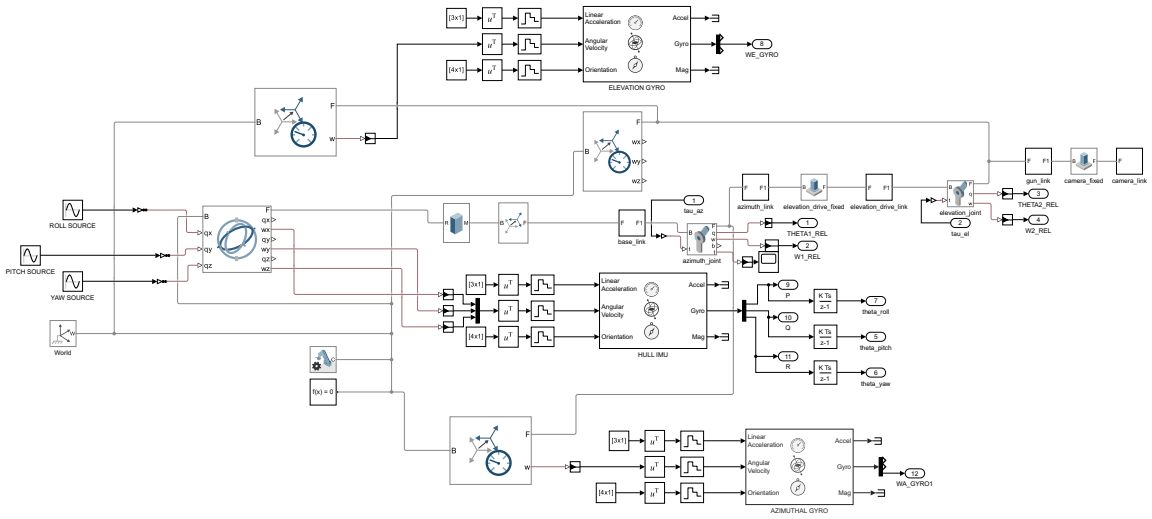


Figure 3. Simulink implementation of the RCWS plant model.

With the weapon maintained at a fixed target angle, there is no commanded target-tracking motion; therefore, the observed pointing error is solely due to vehicle-induced disturbances that propagate through the mechanical system. The resulting pointing error remains below 1 mrad, demonstrating effective disturbance rejection. The disturbance component visible in the corresponding plots represents the residual disturbance that remains after compensation by the proposed control architecture, as shown in Fig. 4.

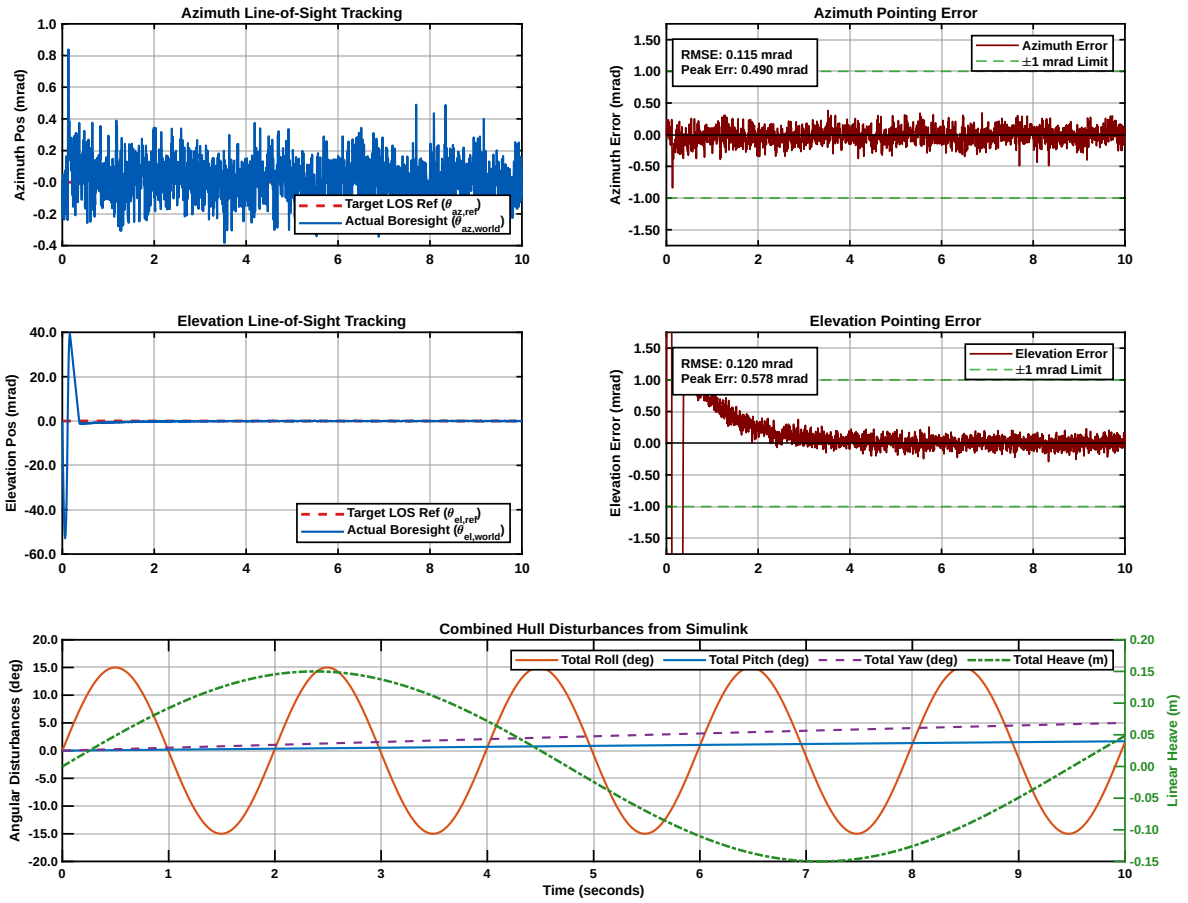


Figure 4. Only Stabilization performance

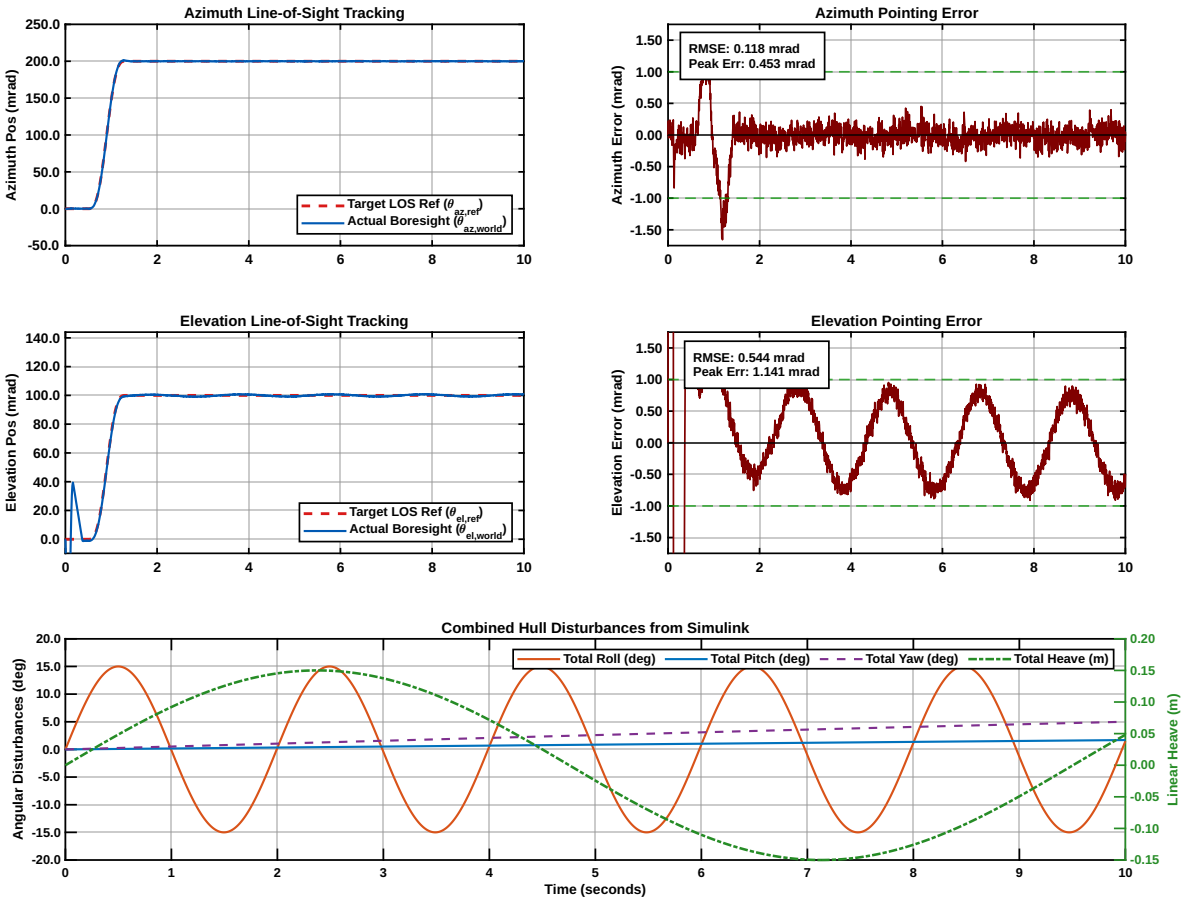


Figure 5. RCWS stabilization and tracking performance.

5. Conclusions

This paper presented a multi-rate control architecture combining Sliding Mode Control, Active Disturbance Rejection Control (SMC-ADRC) and kinematic feedforward compensation for high-precision Line-of-Sight (LOS) stabilization of a two-axis RCWS. By incorporating full spatial cross-axis coupling dynamics into the feedforward path and employing an Extended State Observer (ESO) for active disturbance estimation, the proposed controller effectively attenuates hull-induced motion and compensates for nonlinear system dynamics. Validation on a CAD-derived Simscape Multibody digital twin under multi-frequency base disturbances and tactical-grade MEMS IMU noise demonstrated robust pointing performance, with steady-state RMS errors limited to 0.118 mrad in azimuth and 0.544 mrad in elevation. These results demonstrate the potential of the proposed framework for real-time stabilization of mobile combat platforms. Future work will focus on hardware-in-the-loop (HIL) validation and physical field integration.

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